Reading SP05 · Hull Form Variation from a Parent

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In short: take a good parent hull and change it in two steps. Step 1 sets the shape coefficients; Step 2 sets the size. The sections are never redrawn, so the new hull is as fair as the parent.

	Step 1 · Lackenby	**Step 2 · Stretch**
You choose	C_B (through C_P) and LCB	L, B, T and D
What it does	Slides the stations along the ship, away from midship, Eq. (1); fills the gap at midship with the midship section	Multiplies every offset by a scale factor, Eqs. (3)–(4)
What changes	Station positions, length of parallel middle body, C_B, C_P, LCB, Eq. (2)	The size of the hull
What stays	Section shapes, C_M, the ends (AP, FP, bulb, stern)	C_B, C_M, C_P, C_{WP}, LCB (% L), section shapes, Eq. (5)
Done on	The parent, at its own size	The result of Step 1
Source	Lackenby (1950); Papanikolaou (2026) §4.3, Eqs. 4.17–4.18	Papanikolaou (2026) §4.3, linear distortion (items a1–a2) and Eq. 4.19

Step 1 · Lackenby. In each half of the hull, ξ is the distance of a station from midship as a fraction of the half-length (0 at midship, 1 at the end), and c is the shift factor of that half:

$$\delta\xi = c\,(1-\xi) \tag{1}$$

$$C_{P,\text{new}} = c + (1-c)\,C_P \quad\Longrightarrow\quad c = \frac{\delta C_P}{1-C_P} \tag{2}$$

Step 2 · Stretch. Scale factors, and the new position of every offset point:

$$s_x = \frac{L}{L_0}, \qquad s_y = \frac{B}{B_0}, \qquad s_z = \frac{T}{T_0} \tag{3}$$

$$x' = s_x\,x, \qquad y' = s_y\,y, \qquad z' = \begin{cases} s_z\,z & z \le T_0 \\ T + (z-T_0)\,\dfrac{D-T}{D_0-T_0} & z > T_0 \end{cases} \tag{4}$$

$$C_B = \frac{\nabla}{L\,B\,T} = \frac{s_x s_y s_z\,\nabla_0}{s_x L_0\; s_y B_0\; s_z T_0} = C_{B0} \tag{5}$$

Result: the hull is a function of six design variables:

$$\text{hull} = f\,(C_B,\ \text{LCB},\ L,\ B,\ T,\ D)$$

An optimization model can change any of them and get a new, fair hull in seconds.

Order: always Step 1 first, then Step 2. The stretch keeps whatever coefficients it receives, Eq. (5), so the fullness must be set before it.

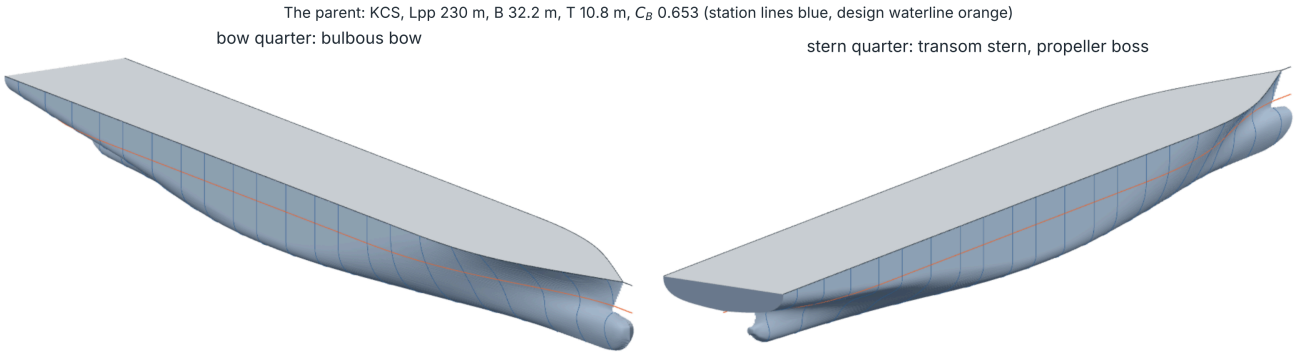
The parent: KRISO Container Ship (KCS)

The KCS is a benchmark container-ship hull designed for 24 kn (F_n 0.26), with a bulbous bow and a transom stern. The course site gives its IGES surfaces and a table of offsets cut from them (Bibliography, *Hull geometry*): half-breadths at 201 stations and every 0.1 m in height, full scale.

Integrated to T, the offsets table reproduces the published particulars:

Quantity	From the offsets	Published
Lpp · B · T [m]	230.0 · 32.2 · 10.8	230.0 · 32.2 · 10.8
Displacement volume ∇ [m³]	52,226	52,030
C_B	0.653	0.6505
C_M	0.985	
C_P	0.663	
C_{WP}	0.833	
LCB [% L, + forward]	−1.50	
Waterline length [m]	232.5	
Forebody C_P · aftbody C_P	0.622 · 0.700	

The 0.4 % difference in volume is small enough for preliminary design.



Step 1 · Lackenby: change C_P and LCB

Each half of the hull is treated on its own. For a station, ξ is its distance from midship as a fraction of the half-length: 0 at midship, 1 at the FP (forebody) or the AP (aftbody). Every station moves away from midship by

$$\delta\xi = c\,(1 - \xi)$$

Midship moves by c, the end does not move, and the gap opened at midship is filled with the midship section. The new prismatic coefficient of that half and the factor needed for a target are

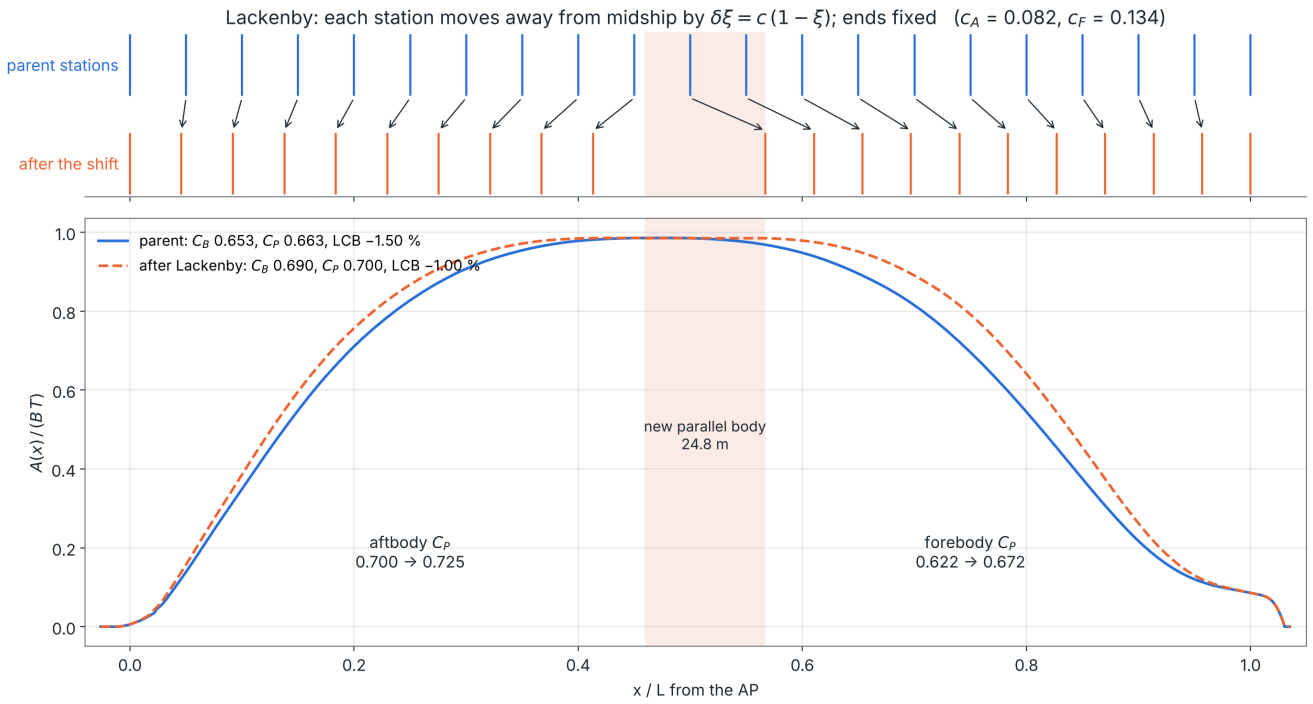
$$C_{P,\text{new}} = c + (1 - c)\,C_P \qquad \Longrightarrow \qquad c = \frac{\delta C_P}{1 - C_P}$$

This is Lackenby's Eq. 4.17 in Papanikolaou (2026) with the new parallel body taken equal to c; the second term of 4.17 then vanishes. Two factors, c_F and c_A, meet two targets, the whole-hull C_P and the LCB: a 2 × 2 solve.

Worked example. Target C_B 0.690 and LCB −1.00 % L, from the KCS (C_B 0.653, LCB −1.50 %). The solve gives c_F = 0.134 and c_A = 0.082.

	Forebody	Aftbody	Whole hull
Parent C_P	0.622	0.700	0.663
Factor c	0.134	0.082	
$c + (1 - c)\, C_P$	0.134 + 0.866 × 0.622 = 0.672	0.082 + 0.918 × 0.700 = 0.725	
Measured after the shift	0.672	0.725	0.700
New parallel body [m]	0.134 × 115 = 15.4	0.082 × 115 = 9.4	24.8 (11 % of L)

The hull is now C_B 0.690, C_P 0.700, LCB −1.00 %, with C_M unchanged at 0.985. A station of the forebody at ξ = 0.5 (x = 172.5 m from the AP) moves to ξ = 0.5 + 0.134 × 0.5 = 0.567, x = 180.2 m: 7.7 m forward. The FP, the bulb and the stern do not move. A finer half needs a negative c, which removes parallel body near midship; the KCS has very little, so it cannot go much finer than the parent.



Step 2 · Stretch: change L, B and T

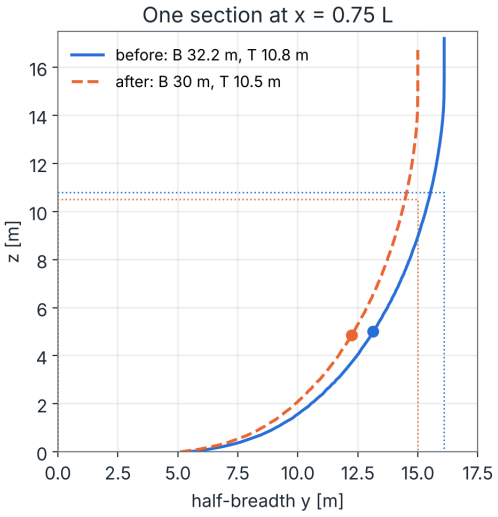
Every offset point is scaled along the three axes:

$$(x,\ y,\ z)\ \rightarrow\ (s_x x,\ s_y y,\ s_z z), \qquad s_x = \frac{L}{L_0}, \quad s_y = \frac{B}{B_0}, \quad s_z = \frac{T}{T_0}$$

Worked example. From the Lackenby hull of Step 1 to an example ship of L 200 m, B 30 m, T 10.5 m (not a project value): s_x = 0.870, s_y = 0.932, s_z = 0.972. L/B goes from 7.14 to 6.67 and B/T from 2.98 to 2.86. One offset point:

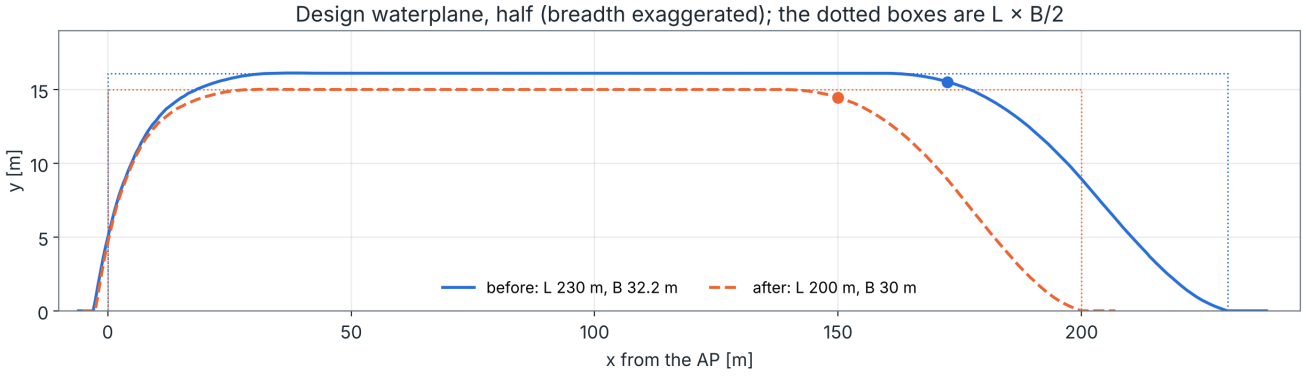
	x from the AP	z above the baseline	Half-breadth y
Before the stretch	172.5 m = 0.750 L	5.0 m = 0.463 T	13.14 m = 0.816 · B/2
After the stretch	150.0 m = 0.750 L	4.86 m = 0.463 T	12.24 m = 0.816 · B/2

Every point keeps its position as a fraction of L, B and T, so the sections keep their shape and only change size.



The marked point: (x, y, z) → (s_x·x, s_y·y, s_z·z)
s_x = 0.870, s_y = 0.932, s_z = 0.972

	before	after	as a fraction
x from the AP	172.5 m	150.0 m	0.750 L
z above the baseline	5.00 m	4.86 m	0.463 T
half-breadth y	13.14 m	12.24 m	0.816 · B/2



Why the coefficients survive. Each quantity scales exactly like the box it is divided by:

Quantity	Scales with	Divided by	Coefficient	Before → after
Volume ∇	$s_x s_y s_z$ = 0.788	LBT	C_B	0.690 → 0.690 (∇ 55,190 → 43,470 m³)
Midship area A_M	$s_y s_z$ = 0.906	BT	C_M	0.985 → 0.985 (342.6 → 310.3 m²)
Waterplane area A_{WP}	$s_x s_y$	LB	C_{WP}	0.854 → 0.854
$\nabla/(A_M L)$			C_P	0.700 → 0.700
Centre of buoyancy x_B	s_x	L	LCB	−1.00 % → −1.00 % (−2.30 m → −2.00 m)

So the two steps do not interfere: Step 1 sets the coefficients, Step 2 sets the size. Stability follows the same scaling (Papanikolaou 2026, Eq. 4.19): BM scales with s_y^2/s_z and KB with s_z.

Depth. Scaling the whole hull by s_z would put the side at $17.2\ \mathrm{m} \times T/T_0$, not at the depth D set by the container grid. The part above the design waterline is therefore stretched on its own, so that the side height of the KCS offsets, D_0 = 17.2 m at midship, lands on D:

$$z' = s_z\, z \quad (z \le T_0), \qquad z' = T + (z - T_0)\,\frac{D - T}{D_0 - T_0} \quad (z > T_0)$$

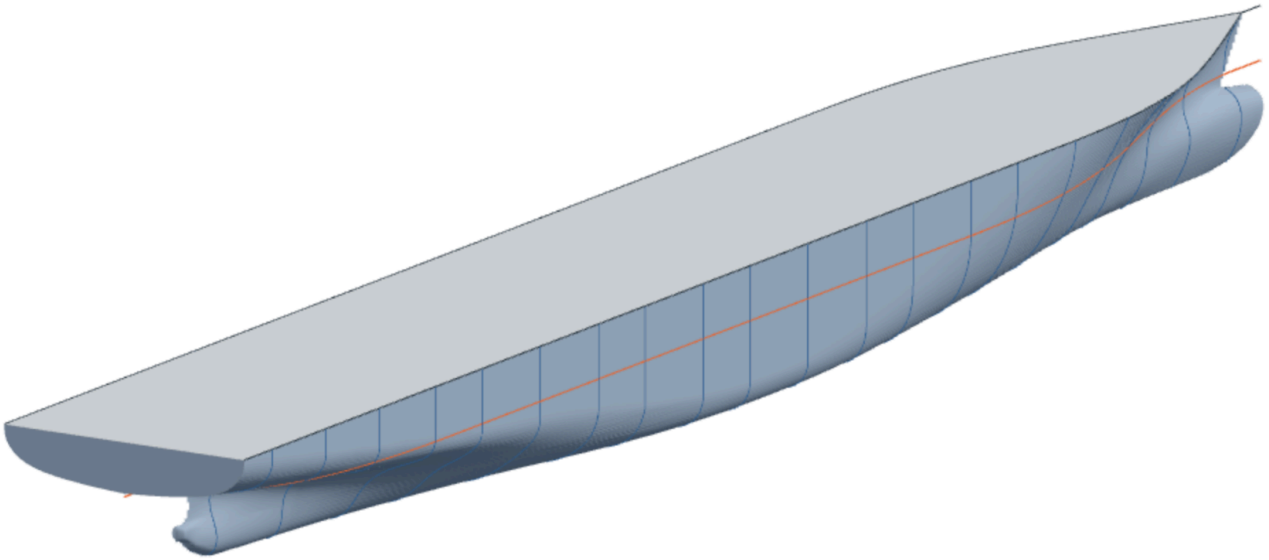
Nothing below the waterline changes, so the hydrostatics at T stay as above; the hull above T, which the bay-by-bay container count needs, reaches D. In the worked example, D = 16.0 m gives a factor of 0.859 above T.

What the two steps do to the hull

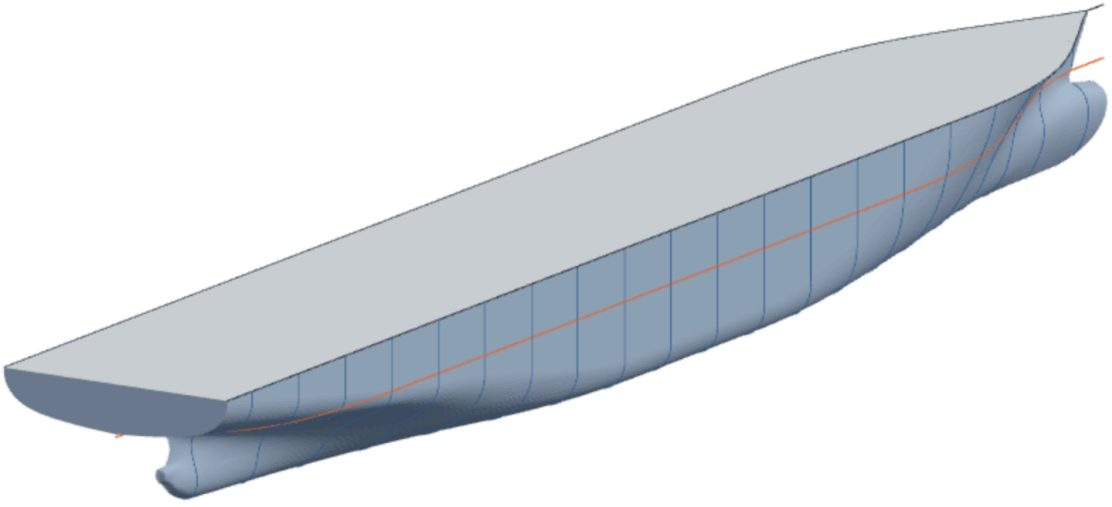
Left: Lackenby adds parallel body at midship and leaves the ends where they were, so the sectional area curve widens in the middle. Right: in non-dimensional form, the stretched variant's sections are the parent's sections at new stations; their shapes are not redrawn.

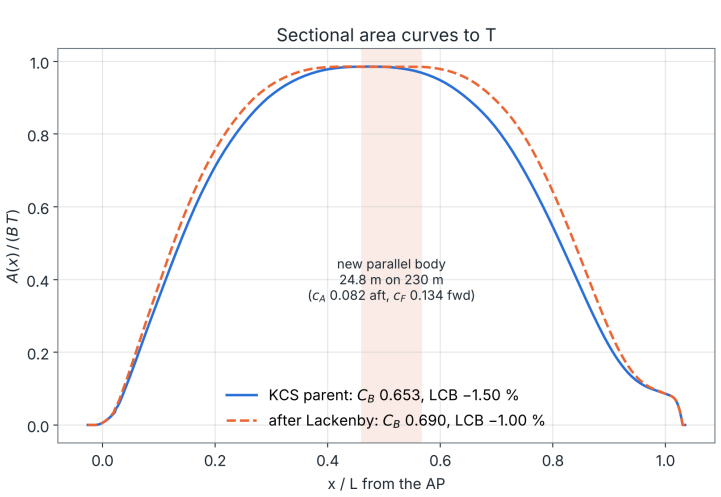
Parent and variant at the same scale

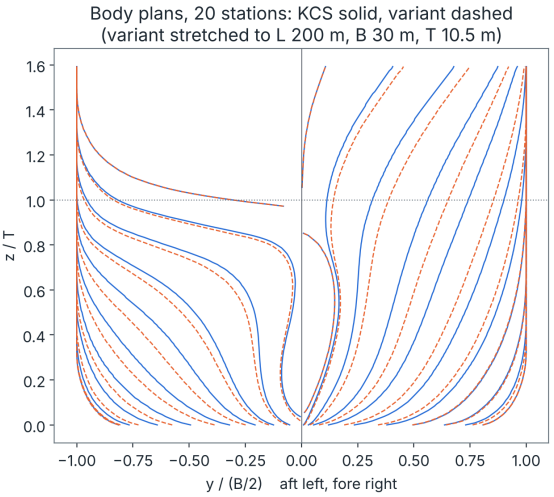
KCS parent: L 230 m, B 32.2 m, T 10.8 m, C_B 0.653, LCB −1.50 %



Variant: Lackenby to C_B 0.690, LCB −1.00 %, stretched to L 200 m, B 30 m, T 10.5 m







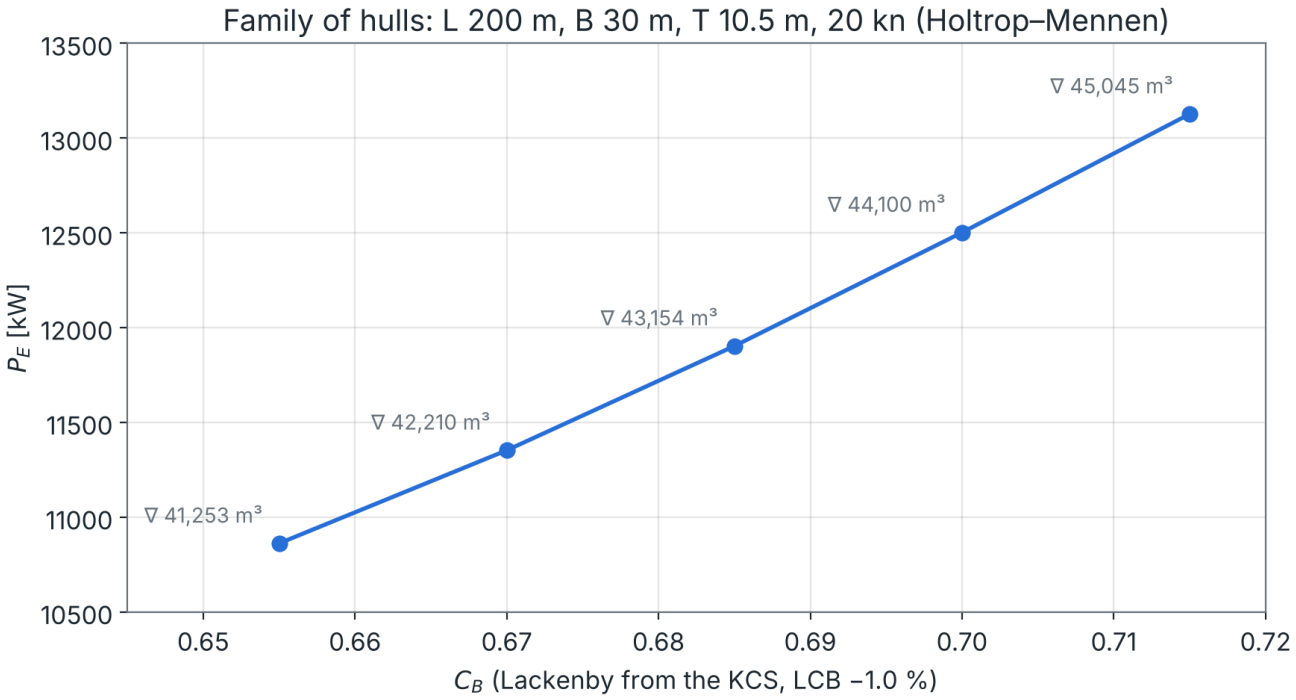
Resistance of the variants, and how far to trust it

The resistance of each variant comes from Holtrop–Mennen (Papanikolaou 2026, §2.14), with the hull-form inputs taken from the hull itself: C_M, C_{WP}, LCB, waterline length, and bulb area and height at the FP. For the example ship at 20 kn (F_n 0.231), the hull gives R_T = 1,176 kN and P_E = 12,098 kW.

On the KCS itself, at model scale (L 7.2786 m, F_r 0.26, Re 1.46 × 10^7), Holtrop–Mennen gives C_T = 3.92 × 10^{-3} against 3.711 × 10^{-3} measured in the towing tank: +5.7 %. The test model had its rudder and was free to sink and trim. Holtrop–Mennen is therefore good for ranking variants against each other; any absolute power taken from it needs a stated margin.

C_B	c_F	c_A	∇ [m³]	R_T [kN]	P_E [kW]
0.655	+0.038	−0.035	41,253	1,056	10,862
0.670	+0.080	+0.014	42,210	1,103	11,353
0.685	+0.122	+0.063	43,154	1,157	11,903
0.700	+0.161	+0.115	44,100	1,215	12,502
0.715	+0.202	+0.165	45,045	1,276	13,127

The table is a family of hulls for the example ship (L 200 m, B 30 m, T 10.5 m, LCB −1.0 %, 20 kn): more C_B carries more displacement and costs more power. Which C_B is best is decided by the rest of the design (deadweight, capacity, cost), not by the hull alone.



Second example: a smaller, slower ship

The same parent, taken to a ship of 150 m at 18.5 kn, with four values of C_B. The inputs:

$$L = 150\ \text{m}, \quad B = 25\ \text{m}, \quad T = 9\ \text{m}, \quad D = 14\ \text{m}, \quad V = 18.5\ \text{kn}$$

Proportions. All inside the recommended ranges:

$$\frac{L}{B} = \frac{150}{25} = 6.00, \qquad \frac{B}{T} = \frac{25}{9} = 2.78, \qquad \frac{D}{T} = \frac{14}{9} = 1.56$$

Scale factors, Eqs. (3)–(4):

$$s_x = \frac{150}{230} = 0.652, \qquad s_y = \frac{25}{32.2} = 0.776, \qquad s_z = \frac{9}{10.8} = 0.833, \qquad \frac{D-T}{D_0 - T_0} = \frac{14-9}{17.2-10.8} = 0.781$$

One hull worked through: C_B = 0.690, LCB = −1.00 % L. Step 1 gives the same factors as in the first example, because Lackenby works on the parent before any stretch:

$$c_F = 0.134, \qquad c_A = 0.082$$

Then the stretch, and the displacement:

$$\nabla = C_B\, L\, B\, T = 0.690 \times 150 \times 25 \times 9 = 23{,}287\ \text{m}^3, \qquad \Delta = 1.025\, \nabla = 23{,}870\ \text{t}$$

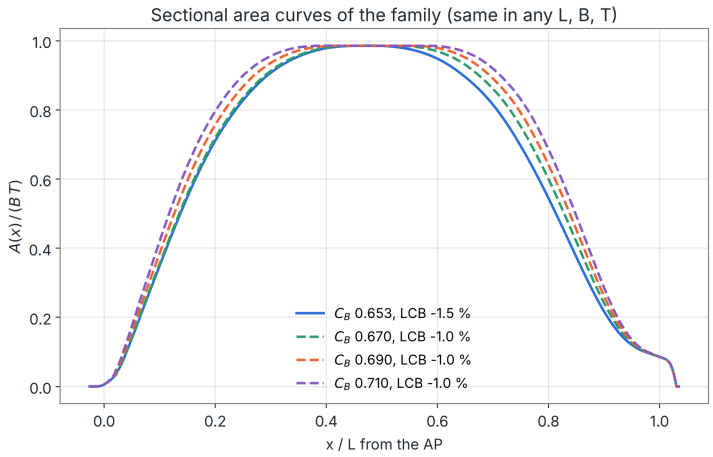
The waterline length of the stretched hull is 152.2 m, so the Froude number is

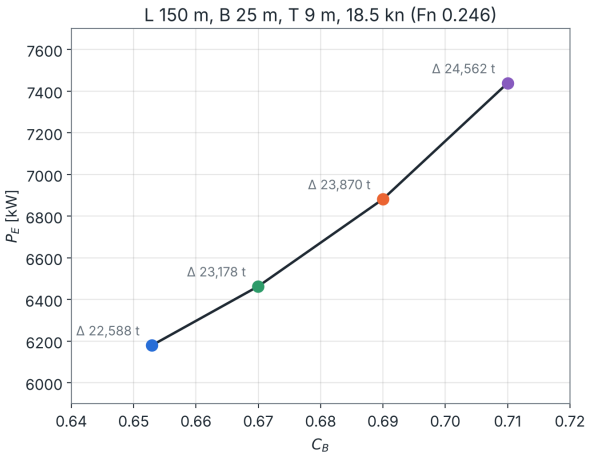
$$F_n = \frac{V}{\sqrt{g\, L_{WL}}} = \frac{18.5 \times 0.5144}{\sqrt{9.81 \times 152.2}} = 0.246$$

The family. C_B 0.653 is the parent's fullness (no Lackenby, LCB −1.50 %); the others are set to LCB −1.00 %. Holtrop–Mennen at 18.5 kn, with each hull's own coefficients:

C_B	c_F	c_A	C_P	C_{WP}	∇ [m³]	Δ [t]	R_T [kN]	P_E [kW]
0.653	0	0	0.663	0.833	22,037	22,588	649	6,180
0.670	+0.080	+0.014	0.680	0.845	22,612	23,178	679	6,463
0.690	+0.134	+0.082	0.700	0.854	23,287	23,870	723	6,881
0.710	+0.189	+0.148	0.721	0.864	23,962	24,562	782	7,438

From C_B 0.653 to 0.710, the displacement grows by 1,974 t (+8.7 %) and the effective power by 1,258 kW (+20 %): the wave resistance grows fastest, from 140 to 235 kN. The coefficients C_P and C_{WP} are the same as in the first example at the same C_B and LCB: they do not depend on L, B, T, Eq. (5).





Apply it: your own hull in five moves

The worked example above, as a procedure. The numbers in brackets are the example's; the notebook cell that does each move is named.

1. **Choose the six hull variables and the speed** (C_B 0.690, LCB −1.00 %, L 200 m, B 30 m, T 10.5 m, D 16.0 m, 20 kn). Check them against the limits below: C_B 0.64–0.71, L/B 5.5–7.5 (6.67), B/T 2.5–3.5 (2.86). *Stretching alone keeps the parent's C_B, 0.653: to get any other C_B, Lackenby comes first.*
2. **Lackenby on the parent** (section 3 of the notebook): solve for c_F and c_A (0.134 and 0.082). Check each half with $c + (1 - c)\, C_P$: forebody 0.622 → 0.672, aftbody 0.700 → 0.725. Read off the new parallel body (24.8 m on the 230 m parent).
3. **Stretch** (section 4): s_x 0.870, s_y 0.932, s_z 0.972. Check that C_B, C_M, C_P, C_{WP} and LCB did not change, and that $\nabla = C_B L B T$ (0.690 × 200 × 30 × 10.5 = 43,470 m³).
4. **Look at it and get its resistance** (sections 5 and 6): sectional area curve and body plan; Holtrop–Mennen with the hull's own C_M 0.985, C_{WP} 0.854, LCB, waterline length and bulb (R_T 1,176 kN, P_E 12,098 kW at 20 kn).
5. **Repeat over a range** (section 7): a family in C_B, or in LCB, L, B or T. Each hull gives ∇ and P_E; your design model adds deadweight, capacity, weights and cost, and chooses.

Limits to state in a report

- **Fullness.** Lackenby fuller is clean up to about C_B 0.71 ($\delta C_P \approx 0.06$, about 17 % of L of new parallel body); beyond it the shoulders become hard. Finer than the parent removes parallel body, and the KCS has very little: below about C_B 0.64 the result is crude.

- **Proportions.** Keep L/B within about 5.5–7.5 and B/T within about 2.5–3.5 (the KCS has 7.14 and 2.98). Further away, the stretched bulb, stern and sections stop being realistic. These ranges are a course guideline, not a rule from a source.
- **Bulb and stern** are stretched with the hull, not redesigned for the new length and speed.
- **Speed.** The KCS was designed for F_n 0.26; the method suits ships near that Froude number.
- **Resistance** comes from Holtrop–Mennen with the hull's own coefficients, not from model tests (+5.7 % on the parent).
- **Cite** the KCS geometry (Tokyo 2015 workshop, NMRI), Lackenby (1950) and Papanikolaou (2026) §4.3.

Support code

Notebook *Demo 8 · Hull Form Variation from a Parent: Lackenby and Stretch* (course site, Notebooks). It loads the KCS offsets from the course site and, with each step a short function: checks the parent's hydrostatics, solves for c_F and c_A, stretches the hull, plots the sectional area curves and body plans, runs Holtrop–Mennen (with the model-scale check above) and builds a family of hulls. Change only the lines marked `# INPUT`.

Sources

- Lackenby, H. (1950). On the systematic geometrical variation of ship forms. *Transactions of the Institution of Naval Architects*, 92, 289 ff.
- Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer. §2.14 (Holtrop–Mennen) and §4.3 (linear distortion, variation of the parallel body, Lackenby method).
- Holtrop, J. and Mennen, G. G. J. (1982). An approximate power prediction method. *International Shipbuilding Progress*, 29(335), 166–170.
- KCS hull geometry (KRISO), Tokyo 2015 Workshop on CFD in Ship Hydrodynamics, National Maritime Research Institute (NMRI), Japan: main particulars and IGES file.
- Measured C_T of the KCS at F_r 0.26 as quoted in *Mathematics* 13(11):1788 (2025).